

BT Series

Data Sheet

USE_8763_M
-BT-BLE Module

Document Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	11/03/2026	AD	AR	BT/BLE Mini Module
1.1	12/03/2026	AD	AK	Added Reset Pin Configuration, RF Performance Recommendation

Table of Contents

Features.....	3
General.....	3
Baseband Features.....	3
RF.....	3
MCU.....	4
Audio Codec DAC.....	4
Audio Codec ADC.....	4
General Description.....	5
Pinout Diagram.....	5
Pin Descriptions.....	6
Digital IO Pin DC Characteristics.....	8
Absolute Maximum Ratings.....	9
Recommended Operating Conditions.....	9
Applications.....	10
Module Diagram.....	10
Module Dimension Drawing.....	11
Recommended Footprint Dimension.....	12
Land Pattern.....	12
Recommended Connections.....	13
Reset Pin Configuration.....	14
Module placement strategy.....	14
Basic guidelines:.....	14
Ordering Information.....	15

Features

General

- Operating voltage: 3V to 4.5V (VBAT)
- Bluetooth 5.3 specification compliant
- Bluetooth classic (BDR/EDR)
- Bluetooth low energy (BLE)
- High speed digital peripheral interface
- UART
- I2C
- I2S
- SPI
- Analog interface
- 4-channel 12-bit ADC
- 8 x PWM I/O
- Smart LED controller
- Selectable GPIO mapping
- Built-in 32Mbits FLASH memory
- Supports OTA firmware upgrade
- GSM 217Hz interference block out design
- Low and Flexible latency

Baseband Features

- 40MHz main clock
- Built-in serial flash for FW storage and parameter upgrade
- Adaptive frequency hopping (AFH)
- Serial Copy Management System (SCMS-T) content protection.

RF

- TX 10dBm (typ.) and 10dBm (typ.) maximal output power for Bluetooth classic and BLE, respectively
- Receive sensitivity (Typical): -94.5dBm (1M BR), -94dBm (2M EDR), and -99.2dBm (1M BLE) and -95dBm (2M BLE).

MCU

- Realtek Real-M300, 32-bit ARMv8M mainline processor without floating point unit (FPU)
- Memory Protect Unit (MPU)
- 4MB Flash
- 532KB embedded shared SRAM
- 40MHz, 80MHz, or 100MHz operation

Audio Codec DAC

- **Class-D Output Configuration** → *BD type (inductor-free), differential output*
- **Supply Voltage (AVCC_DRV)** → 1.8 V (± 0.1 V tolerance)
- **Resolution** → 24-bit DAC
- **Sample Rates Supported** → 8 to 96 kHz
- **SNR (Signal-to-Noise Ratio)** → 106 dBA (32 Ω) / 104 dBA (16 Ω)
- **THD+N (Distortion)** → -85 dB @ 40 mW, 16 Ω load
- **Crosstalk** → -95 dB (excellent channel isolation)
- **Noise Floor** → -106 dBA (very clean output)
- **Output Power (typical):**
 - 31 mW @ 32 Ω load
 - 40 mW @ 16 Ω load
- **Allowed Load** →
 - Resistive: $\geq 16 \Omega$
 - Capacitive: ≤ 100 pF

Audio Codec ADC

- **Resolution:** 24-bit
- **Input Sample Rate (fs):** 8 kHz – 96 kHz
- **Analog Supply Voltage:**
 - AVCC_DRV: 1.7 – 2.2 V (Typ. 1.8 V)
 - AVCC: 1.7 – 2.1 V (Typ. 1.8 V)
- **ADC Channels:** 3
- **MIC Boost Gain:** 0 – 30 dB (steps: 0/3/6/9/12/18/24/30 dB)
- **Maximum MIC Input Level:**
 - Differential: -30 dBV (at 30 dB boost), 0.5 dBA (at 0 dB boost)
 - Single-ended: -36 dBV (at 30 dB boost), -6 dB (at 0 dB boost)
- **Line Input Maximum Level:** -6 dBV (single-ended, 0 dB gain)

General Description

The USE_8763_M is a high-performance, compact, low-power Bluetooth module. It supports stereo audio output, multiple microphone inputs and UART communication, and an advanced PMU design, making it suitable for Bluetooth audio receiver and transmitter applications.

For more information reach out to us at info@unisemindia.com.

Pinout Diagram

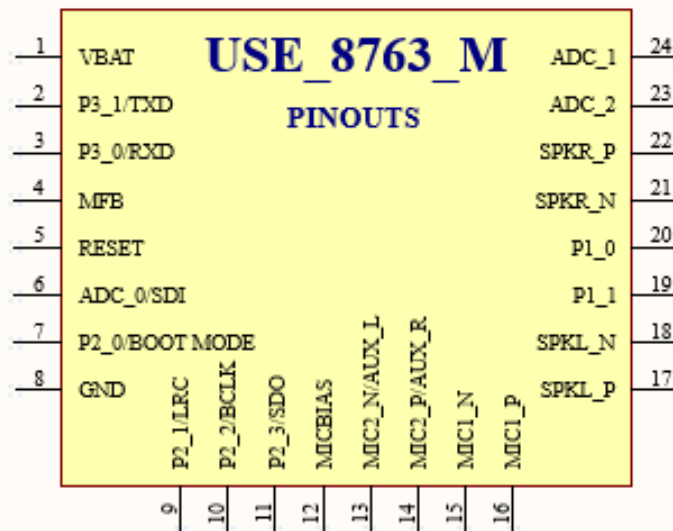


Fig 1 : Module Pinouts

Pin Descriptions

Pin No.	Pin Name	Type	Function Description
1	VBAT	PIO	Battery input when battery only. Battery charge output when in charger mode with adapter in.
2	P3_1/TXD	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_HV33 power domain UART_TX for Flash memory programing
3	P3_0/RXD	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_HV33 power domain UART_RX for Flash memory programing
4	MFB	I_PU	Multi-function button input with internal pull high, low active with at least 3ms low.
5	RESET	I_PU	System reset input with internal pull high, low active with at least 8ms low to trigger system reset
6	ADC_0/SDI	I/O I/O A	Programmable GPIO. Programmable as ADC input pin Configurable pull high/low state LDO_AUX power domain
7	P2_0/BOOT MODE	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_AUX power domain HCI mode selection. H: APP mode L: HCI mode
8	GND	GND	Exposed pad with ground connections
9	P2_1/LRC	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_AUX power domain
10	P2_2/BCLK	I/O PU	Programmable GPIO. Configurable pull high/low state

			LDO_AUX power domain
11	P2_3/SDO	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_AUX power domain
12	MICBIAS	PO, AH	Microphone bias output
13	MIC2_N/AUX_L	AH	MIC2 input negative pad. Used as 2nd MIC in dual MIC application. AUX input left channel pad.
14	MIC2_P/AUX_R	AH	MIC2 input positive pad. Used as 2nd MIC in dual MIC application. AUX input right channel pad.
15	MIC1_N	AH	MIC1 input negative pad. Used as main MIC in dual MIC application
16	MIC1_P	AH	MIC1 input positive pad. Used as main MIC in dual MIC application
17	SPKL_P	AH	Left channel speaker output positive
18	SPKL_N	AH	Left channel speaker output negative
19	P1_1	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_HV33 power domain Serial wire Debug Clock
20	P1_0	I/O PU	Programmable GPIO. Configurable pull high/low state LDO_HV33 power domain Serial wire Debug Input and Output
21	SPKR_N	AH	Right channel speaker output negative.
22	SPKR_P	AH	Right channel speaker output positive.
23	ADC_2	I/O I/O A	Programmable GPIO. Programmable as ADC input pin Configurable pull high/low state LDO_AUX power domain

24	ADC_1	I/O I/O A	Programmable GPIO. Programmable as ADC input pin Configurable pull high/low state LDO_AUX power domain
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Pad Type	Definition
I/O	Bidirectional digital pad
I/O PU	Bidirectional digital pad with pull high resistor inside when input mode
AI/O	Analog input and output
I_PU	Input with internal pull high inside
PI	Power Input
PO	Power output
PIO	Power Input and Output
I/O A	Bidirectional digital pad and programmable ADC
AH	Analog and digital hybrid programmable

Digital IO Pin DC Characteristics

Item	Min	Typ.	Max	Unit
Input low voltage (VIL)	-0.4	-	0.4	V
Input high voltage (VIH)	0.7 X VDDIO	-	VDDIO+0.4	V
Output low voltage (VDDIO=1.8V)	-	-	0.2	V
Output high voltage (VDDIO=1.8V)	VDDIO-0.2	-	-	V
Output low voltage (VDDIO=3V)	-	-	0.4	V
Output high voltage (VDDIO=3V)	VDDIO-0.4	-	-	V

Absolute Maximum Ratings

Item		Min	Typ.	Max	Unit
Storage Temperature		-40	-	+105	°C
Power management Unit					
VDD_CORE	VDDCORE	-	-	1.32	V
Synthesizer control	VD12_SYN	-	-	1.32	V
RF transceiver control	VD12_RTX	-	-	1.32	V
Codec	AVCC	-	-	2.2	V
Headphone driver	AVCC_DRV	-	-	2.2	V
VDDIO0	HV33	-	-	3.6	V
VDDIO1/ADC power	VDD AUX	-	-	3.6	V
BAT IN	VBAT	-	-	5.5	V

Recommended Operating Conditions

Item		Min	Typ.	Max	Unit
Operating Temperature	RTL8763ESE	-30	25	85	°C
	RTL8763EAV-VI	-40	25	105	°C
Power Management Unit					
VDD_CORE	VDDCORE	1.14	1.2	1.26	
Synthesizer control	VD12_SYN	1.14	1.2	1.26	V
RF transceiver control	VD12_RTX	1.14	1.2	1.26	V
Codec	AVCC	1.7	1.8	2.1	V
Headphone driver	AVCC_DRV	1.7	1.8	2.1	V

VDDIO0 power	HV33	3.0	-	3.6	V
VDDIO1/ADC power	VDD_AUX	1.8	-	3.6	V
BAT IN	VBAT	2.8	-	4.5	V

Applications

- Bluetooth audio transmitter
- Bluetooth audio receiver

Module Diagram

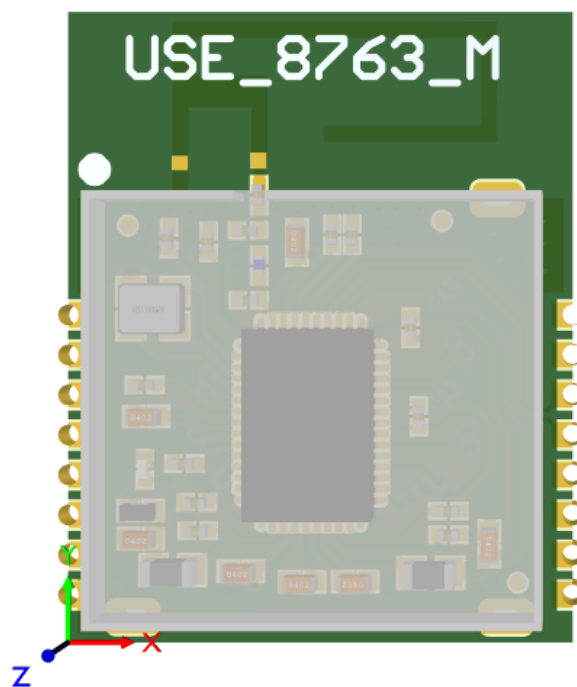


Fig 2(a) : Module with shield (Top view)

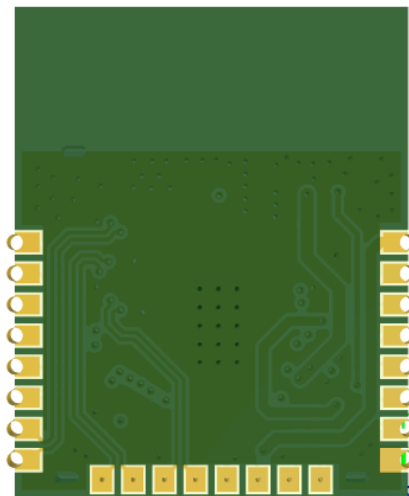


Fig 2(b) : Module (Bottom view)

Module Dimension Drawing

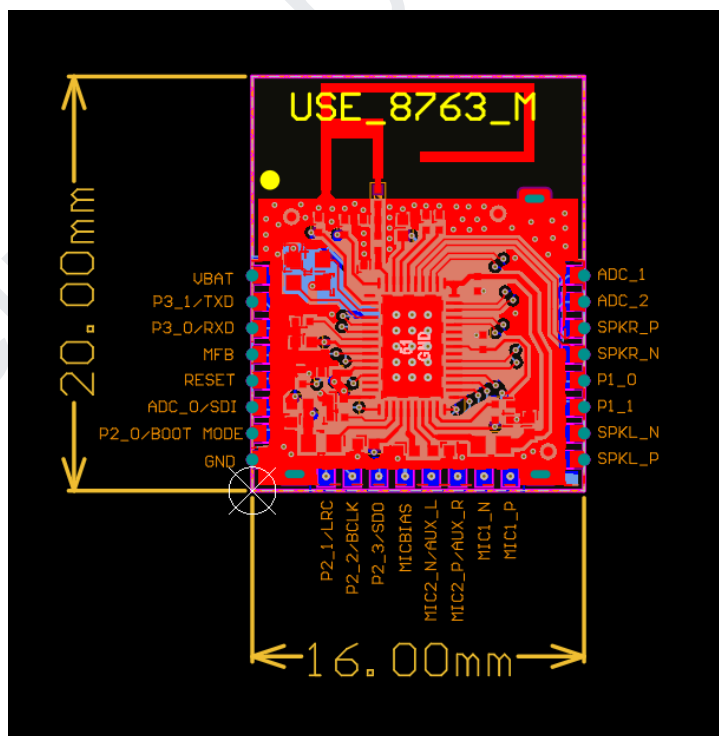


Fig 3 : Module pinouts & Dimension

Recommended Footprint Dimension

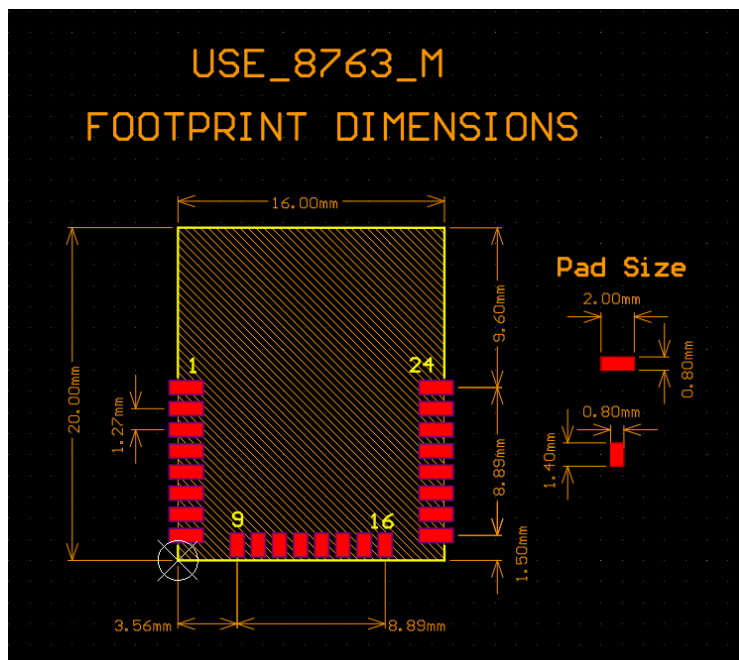


Fig 4 : Module Footprint dimension

Land Pattern

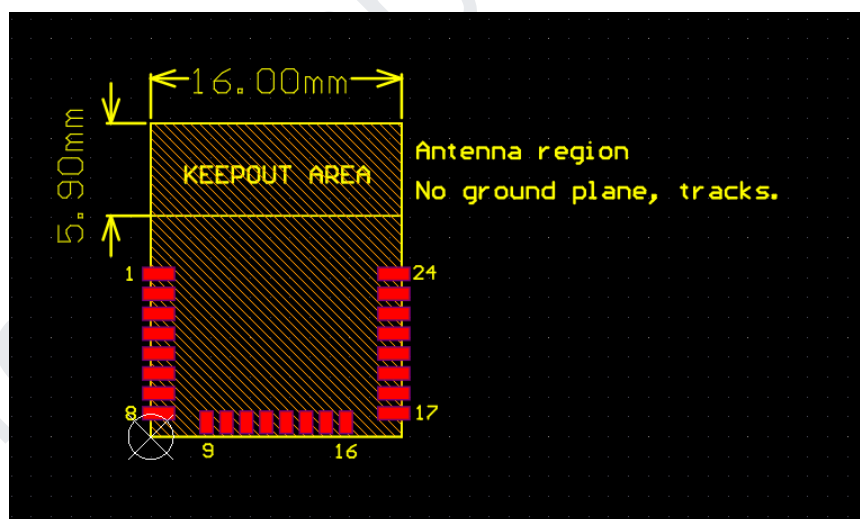


Fig 5(a) : Standard Recommended Footprint

- As illustrated in Fig. 5(a), the antenna keep-out region shall be kept completely free of copper on all PCB layers, including ground planes, signal traces, and vias, to ensure proper RF performance.

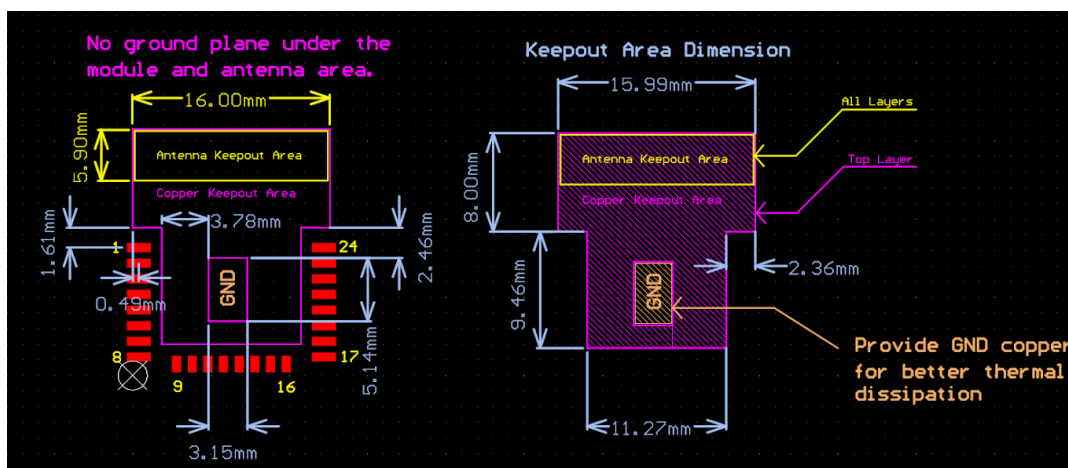


Fig 5(b) : Preferred Footprint

1. Copper Keepout Area

As illustrated in Fig. 5(b), ensure the Copper keepout area is completely free of copper and signal traces on the top layer of the carrier board.

2. Antenna Keepout Area

As illustrated in Fig. 5(b), the antenna keepout region must remain completely free of copper on all PCB layers.

Note : Use the footprint recommendations shown in Fig. 5(a) or Fig. 5(b), as appropriate for the target application.

Recommended Connections

The diagram (shown below Fig 6.) of the module application on external filter parts, when needed, please design in the mainboard, the parts parameter can depend on the actual need to change. And pay attention to the increase and decrease ramping of supply voltage faster than 0.5v/ms;

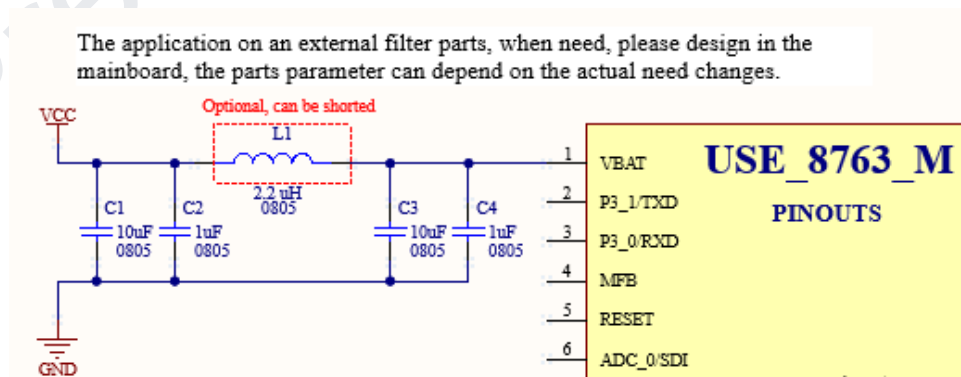


Fig 6

Reset Pin Configuration

The Reset pin has an internal pull-up resistor to VDD. The module has a 1uF capacitor built-in, connected to the Reset pin. An additional capacitor $>0.1\mu\text{F}$ may be added depending on application requirement without affecting the timing requirements.

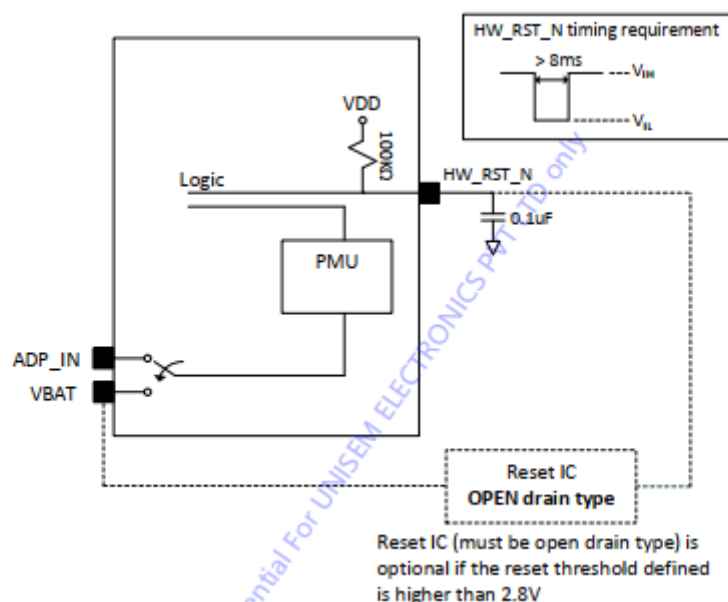


Fig 7

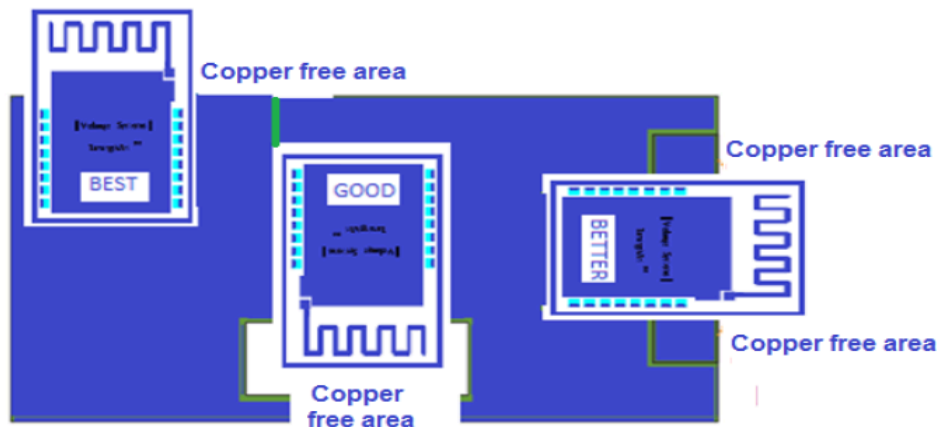
Module placement strategy

BLE/Sub-1GHz modules are RF devices that require proper placement on PCB to ensure optimal performance. The antenna on the PCB has (about) an omnidirectional radiation pattern. To maximize antenna efficiency, an adequate grounding plane must be provided under the module. However, the areas underneath and surrounding the antenna area must be free of copper. The position of the module on the host board and overall design of the product enclosure contribute to antenna performance. Poor design affects radiation patterns and can result in reflection, diffraction, and/or scattering of the transmitted signal thus limiting the range.

Basic guidelines:

- Never place the ground plane or route copper traces directly underneath the antenna portion of the module.
- Never place the antenna close to metallic objects.
- Keep wiring, components and objects away from the antenna.
- Do not place the antenna in a metallic or metalized plastic enclosure.

- Enclosure walls should be 1cm or more away from the antenna in all directions.
- If possible, mount the antenna overhanging the edge of the host board (Motherboard). Add an uninterrupted ground plane on the host board, directly underneath the top shielded zone of the module, up to the PCB edge. Adding a ground plane will allow traces to be run on the bottom side of the host board if required.
- If the antenna cannot be mounted in an overhanging position, then provisions must be made to keep the area clear of copper.



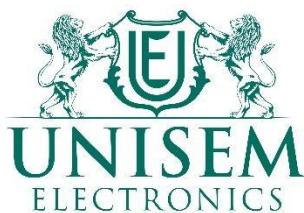
Module Placement Recommendation

Fig 8

Ordering Information

Ordering Part number	Chipset	Description
USE8763MI	RTL8763ESE	-30 to 85°C
USE8763MA	RTL8763EAV-VI	-40 to 105°C

For ordering, please contact us on info@unisemindia.com.



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